

1 Barcode Input

A barcode is machine-readable information, which may also directly be generated within the shop floor system. The principal reading devices used are: barcode scanners, swipe card readers or scanners. By default, the terminal supports the barcodes "39", "128" as well as "interleaved 2 of 5 "



Only barcode readers connected to the serial interface (COM interface) can identify barcodes and assign them automatically to the respective input fields at the terminal. This is impossible for barcode readers "connected" via keyboard.

Barcode structure for operations

AAAAAAAAAAAAAGGGGG

Place	Name	Min. length	Max. length *	Example
*	Asterisk	1	1	*
A	Order number	1	12	8
F	Sequence number	0	2	0
G	Operation number	1	4	3
S	Split number	0	2	0
*	Asterisk	1	1	*
Length of BDE order barcode without asterisk (example)				11

* Depends on the configuration of field lengths in HYDRA's basic parameter settings

Barcode structure for machines/workplaces

MMMMMMMMM

Place	Name	Min. length	Max. length	Example
*	Asterisk	1	1	*
M	Workplace/ machine number	8	8	8
*	Asterisk	1	1	*

Place	Name	Min. length	Max. length	Example
Length of machine/workplace barcodes without asterisks				8



To identify a barcode as a machine/workplace number, the barcode must be 8 characters long. If the machine/workplace number is set to "numerical" in the basic parameter settings, it has to be filled up with leading zeroes to reach 8 characters. In case of alphanumeric machine/workplace numbers, they have to be filled up with underscore characters ("_") to the right.

Barcode structure for machine statuses

NNNNN0

Place	Name	Length
*	Asterisk	1
N	Machine status, with leading zeroes	5
0	Always "0"	1
*	Asterisk	1
Length of status barcodes without asterisks		6

2 AIP configuration of barcodes

2.1 Configuration in ctaip.ini

Readers are configured in section [comports] of the file ctaip.ini:

Section [comports]	
COM1=0 COM2=0 COM3=0 COM4=0 COM5=0 COM6=0	Possible initializations of comports COMx=0 => is not used (by default) COMx=BAR COMx=PSTD PLEASE NOTE! Only starting from reader firmware 69355E COMx=LEGIC COMx=PLG COMx=RFLESER COMx=MSS COMx=SLGB Entry for readers of the type "Schlagbaum" COMx=UKEY U-Key readers Byte 14 and 15 set up the badge/ID card number e.g.: 01010D01 0E020000 04000003 E7175600 03E7 → badge number → 999 1756 → company number → 5974 COMx=KABALEG Kaba Benzing Legic With Bedanet 9580 always COM4 7 bytes are transferred as of byte 15 XXXXXXXXXXXXKKKKK F:= company number K=badge number COMx=MBB-S6 MBB-S6 reader Attention! Please note! This reader type requires the RS-485 converter to be modified (ECHO=OFF RTS=High) Comports masking has to be set for CLEA + DEUT MBB-S6-MASK=XXXXXXXXXXXXKKKKK COMx=DRV_UCR New LEGIC Advant PZE/MF reader. required for new PZE readers. The names of these readers include "LGA". Please note: If badges/ID cards are used <u>not</u> complying with MPDV's standard ID, the following parameter must be used for masking in section [Comports-Mask]: DRV_UCR-MASK=.... COMx=PLG-CRYPT New LEGIC Advant ZKS reader required for new ZKS readers. The names of these readers include "LGA". It might be necessary to customize the file plg_crypt.ini for specific customers Set this parameter if Kabalegic and badges/ID cards with MPDV ID are used KABALEGM=ON By default, the Kaba Benzing badge ID is used. In this case the

	parameter is to be commented out As of ctaip #V 2.0.2.27 Read out Kabalegic using configurable search string (e.g. 4F) KABALEG-SEARCHSTRING=XX Correct configuration takes priority (2 places, values 0..9,A..F)
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Section [Comports-Mask]	Example
LESERTYP+'-MASK' =Masking	applicable abbreviations for masking 'T' = Telegram number 'L' = Reader number 'F' = Company number 'K' = Badge/ID card number 'E' = Replacement number 'P' = Check digit (not implemented) otherwise e.g. 'X' = Placeholder (character to be ignored)
Examples: SLGB-MASK=TTTTLLFFFFFFFKKKK BAR-MASK= MBB-S6-MASK=XXXXXXXXXXKKKKKK	Masking for readers of the type "Schlagbaum": TTTTLLFFFFFFFKKKK = Data string of the reader The badge number may be recorded at every position of the data string Please note!! Impossible for PLG // Status 15 October 2003 DB

The below section has been designed for the configuration of (PLG) Polling Legic Readers or (PSTD) Polling Swipe Card Readers. These readers are mostly used for PZE/ZKS terminals (CT-380) or ZKS terminals (CT-385).

Section [init]	
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PlgTimeOut=2	→ TimeOut when starting polling
FreischaltungZyklus=3	→ Cycle for activating/releasing access
MaxComError=15	→ Maximum number of communication errors until access is in "malfunction"
UngueltigZyklus=300	→ Cycle for sending the status "invalid" (invalid badge) in order to re-initialize the reader
InitComError=50	→ Number of communication errors in order to re-initialize Comport
MSSImmediateWrite=false	→ ZKS immediately activates channels for MSS connection (true = not efficient)
Leser-4-Hupe-Aktiv=true	→ Enables an acoustic signal (buzzer/horn) for the status "valid" (valid badge)
LnrErrorPCnt=100	→ Polling is only performed with every n-th attempt if the reader is in the status "malfunction=e". (priority polling)
PinCodeLen=4	→ Length of PIN code input
ComResetTimer=5	→ Minimum cycle for comport re-initialization
ReInitComport=true	Activation of automatic comport re-initialization → by default = false (= without automatic re-initialization)

The below section has been designed to configure MBB readers triggered via a thread in a DLL. These readers are only used for ZKS terminals (CT-385).

Section [MBB-S6-configuration]	
MinMSecDauerOeffner=200	→ MinMSecDauerOeffner = Minimum duration "door opener" in MSec -> by default [200]
*****	*** MBB-S6 - system settings (do not change) *****
ProcessMessages=true	→ ProcessMessages = Processing of Windows messages while waiting for the application result -> by default [true]
ThreadSleep=20	→ ThreadSleep = Sleep in MSec of MBB-DLL while waiting for application result -> by default [20] range of values [10 .. X] -> changes might lead to increased CPU loads or bad response times

2.2 Configurations in hytnrcfg.ini

Entry	Comment
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Section [Terminal->USR 0]	
SuppressBarcodeError=On	Suppresses messages such as 'Barcode ... is wrong', etc. Required if the barcode is processed in the script and normal identification processes cannot identify the barcode
OnBarcode=P_AN	If the AIP basic screen is opened and a barcode is read in, the dialog P_AN opens instead of the dialog M_MST. The scanned badge/ID card number is transferred.
BarcodePrefixChar=\$	Configuration of an alternative separator for barcode prefixes. Can be used if a dot not used as prefix identifier is included at the third place of a real barcode.
ON-KNR-CODE=<Dialog> (By default "M_INFO.PERS")	As of V# 2.0.2.57 Configures the dialog that opens from the main dialog when scanning the staff badge/ID card number using readers such as Legic, Kabaleg, UKey, .

2.2.1 Notes/configurations for concurrent lengths

Entry	Comment
Section [terminal configuration 0] and/or [terminal configuration 2XXX];	(general configuration) (2XXX terminal-specific configuration)
ANR-COMPLETE-BARCODE-ONLY=TRUE	Option to avoid processing of incomplete order barcodes. (e.g. AUNR,AUNR+AFOLG,...) -> The option is set to <FALSE> by default
RIVAL-BARCODE-LEN-INFO-MSG=FALSE	Option disabling messages indicating that barcodes cannot be processed due to concurrent lengths. -> The option is set to <TRUE> by default

Please note for configuration: the section may only be inserted provided it does not yet exist.

Sample configuration (for all terminals)

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[Tnr configuration 0]
ANR-COMPLETE-BARCODE-ONLY=TRUE
BARCODE-LEN-INFO-MSG=TRUE
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The above configuration prevents incomplete order barcodes from being processed at all terminals (configuration for one terminal with [TNR configuration 2xxx] xxx=terminal number including leading zeroes "0"). If this option is set, only complete order barcodes will be processed. Consequently, valid lengths for order barcodes result from added partial order lengths (Please note: //- ANR (order number) parts with (*) may have the length '0')

- ➔ AUNR + AFOLG (*) + AGNR + UAGNR (*)
- ➔ AUNR + AFOLG (*) + AGNR + UAGNR (*) + SPLNR (*)

The following message is shown if a barcode cannot be processed due to concurrent lengths:

Note [concurrent length]

A barcode (<VAL>)<n>whose field assignment (<IDS>)<n>cannot clearly be defined, has been scanned.<n><n>Processing is only possible<n>if one of the fields indicated is focused.